

The Common Operating Ecosystem for Supply Chain Intelligence with **Physical AI**

Forecast demand. Rebalance stock. Reduce waste. Bidding to Buy.

Every site forecasts alone, orders alone, and writes off alone.

There is no shared picture of what the business is holding — so one site buys what another is days away from discarding.

4–10%

of everything purchased, written off

\$72,000

average waste, per site, per year

\$162B

a year, in US food service alone

THE PRIMITIVE

One picture. One decision.

01

Forecast each branch

Demand per item, exploded through the bill-of-materials into what each establishment will actually consume.

02

Rebalance across the franchise

A shortage at one branch checks every other branch for surplus — nearest establishment, nearest expiry — before it checks a supplier.

03

Buy only the remainder

Only the net shortage across the whole group reaches a supplier — one purchase order, one human approval.

WHAT IT READS — ALL OF IT ALREADY EXISTS IN THE BUSINESS

Sales

POS API — Toast, Square, Lightspeed

Recipes

Bill-of-materials, entered once

Prices & receipts

Supplier invoices, distributor EDI

On hand

Their existing counts, with a confidence band

No new hardware, no new process. The audit that opens the sale runs on historical exports alone.

Four sources in. One decision out. One tap back.

READS — ALL ALREADY IN THE
BUSINESS

Sales

POS API — Toast, Square, Lightspeed

Recipes

Bill-of-materials, entered once

Prices & receipts

Supplier invoices, distributor EDI

On hand

Their existing counts, with a confidence band

CROSSHAUL DECIDES

- 1 Forecast demand per branch
- 2 Explode through the bill-of-materials
- 3 Build one franchise-wide position
- 4 Match surplus to shortage — nearest branch, nearest expiry
- 5 Tender only the remainder

ACTS — A HUMAN APPROVES, ALWAYS

Transfer proposed

Donor site gets one accept or decline

Purchase order drafted

Owner releases it through their own channel

Dated stock flagged

Cleared through demand, not written off

And it calls you

The decision arrives as a phone call, not a tab

Branches never contact each other — that is the point. No new hardware, no new process, and the opening audit needs none of it live.

LIVE, RIGHT NOW

One shortage in. One decision out.

Downtown is **36 kg short** on tomatoes going into the weekend. Marina holds 34 kg — surplus, expiring in two days. Mission is itself below par, so it cannot donate. CrossHaul moves 10 kg from Marina, puts the remaining 26 out to tender, and Bay Foods wins at \$2.05 against the incumbent's \$2.20. The owner gets a phone call, not a dashboard.

Purchase avoided	+\$20.50
Spoilage avoided	+\$20.50
Tender won	+\$3.90
Courier leg	-\$4.00
Net, one decision	+\$40.90

Why this was not built five years ago

Coordination got cheap

Per-site agents negotiating continuously only became affordable in the last eighteen months. Before that, cross-site planning meant an enterprise ERP that only fifty-unit chains could buy.

Margins stopped forgiving waste

Food and labour costs since 2024 mean operators will now pay for single-digit-percent recovery they used to write off without looking.

Operators are crossing into multi-site

3,381 locations are moving through the one-to-nineteen unit transitions in 2026 alone. The day they open a third site, they acquire a problem they have never had.

And nobody sells them a tool for it — which is the next slide.

The market, derived two ways

CATEGORY TAM

\$4.55B → \$9.18B

Restaurant inventory & purchasing software, 2025 to 2030. 15% CAGR.

US SAM

\$672M / yr

225,000 multi-site locations at \$2,988 a year.

BEACHHEAD

3-20 units

63.5% of multi-concept companies. Median operator: 13 units.

Checked bottom-up, independently

US restaurant locations	750,000
× price per month	\$249
× months	12
US alone	\$2.24B

A US-only bottom-up at roughly half the global category is internally consistent. Two methods agreeing is the point — the number is not copied off an analyst chart.

They buy a 3.6× return

\$249

per location, per month

Waste exposure per location	\$72,000 / yr
Recovered, at our assumed 15%	\$10,800 / yr
What CrossHaul costs	\$2,988 / yr
Return on the line item	3.6×

The 15% is deliberately low, and it is the number I would defend hardest. We cannot touch over-portioning, theft or prep loss — only the spoilage and stockout component that shows up in inventory data. Its smallness is the credibility.

Three years, built bottom-up

	Year 1	Year 2	Year 3
Groups	16	58	167
Locations	96	361	1,083
Ending ARR	\$288K	\$1.08M	\$3.24M
Gross margin	76%	81%	84%

4.0 mo

CAC payback

7.5×

LTV / CAC, capped at 36 months

-\$73K

Year-3 EBITDA before growth spend

Uncapped, LTV/CAC is over eighteen — a number nobody believes, so I cut it to three years. And year three loses money only because we are buying customers that repay in four months. Strip that out and the installed base almost pays for the company.

Monthly cohort model — every figure derives from one inputs block. github.com/abhijitbetigeri/CrossHaul

Food service is the beachhead, not the ceiling

Sector	Status
Food service	Built. Live. Won.
FMCG manufacturing	Thesis — not built
Grocery & convenience	Thesis — not built
Pharmacy	Thesis — not built

SAP has run inter-plant stock transfers for decades. A six-plant manufacturer does not run SAP. **The mid-market of every sector has the same problem and the same non-solution — that is one market, not five.**

Where physical AI actually sits

A decision is only worth what it costs to execute. We reason about vehicles, distance and spare capacity — and we have built the simulation where a robot runs the transfer end to end, emitting the task record.

Playable simulation

abhijitbetigeri.github.io/Project-SCIM/sim/

source — github.com/abhijitbetigeri/Project-SCIM

We are a **beneficiary** of transport autonomy, not a producer of it.

When a ten-unit city leg costs \$2 instead of \$7, every marginal transfer turns positive.

At 1,083 locations: 931 tons kept out of landfill, ~4,740 tonnes CO2e — the consequence, not the pitch

Both are load-bearing, neither is decoration

ElevenLabs — the approval channel

The failure mode of this category is not features. It is that the person who owns the stock is on their feet all day and never opens a dashboard. So CrossHaul **calls them**, states the decision it already made, and takes "yes" for an answer. A published conversational agent is embedded in the product; text-to-speech renders the same call to disk so the demo cannot die on venue wifi.

k3trn3a2.insforge.site/voice.html

Lower adoption cost is a go-to-market argument, not just a product one.

Nebius — routing by reasoning

Sending every call to a frontier model is the same mistake as running every query on your largest database instance.

Forecasting and bulk extraction run on open-weights models; the multi-constraint rebalance stays frontier

Inference, unrouted	\$5.16
Inference, routed	\$0.88
Removed, per location per month	83%

Verified live: the forecast call costs **79× less** and the decision that spends money is untouched.

Honest limit: inference is not the dominant cost line — customer success is. Routing is about a fifth of the modelled \$60 → \$ reduction.

CrossHaul is the routing layer for inventory Management

The system that decides to have agentic Intelligence in Supply Chain ecosystem

\$1.5M

pre-seed

Burn to month 24 is \$772K. This covers it with a year of buffer.

34 mo

runway on it

The month cumulative burn crosses \$1.5M in the cohort model.

\$1.08M

ARR at month 24

58 groups × 6.2 locations × \$2,988 — the Series Seed milestone.

n

pilots to sign first

Clustered Bay Area groups — and the audits that test the whole thesis.